Experiments

17 task(s) across 13 run(s) · generated 2026-08-21.

Fixed-input tests

arm	comparison	calls	total input	total output	realized reuse	median TTFT	p95 TTFT	median TPOT	median request latency	wall
A_c8_w4 · 08-18 15:34										
c8w4 hit miss 20260820 115248	open comparison									
miss · 08-20 13:29		108	816,288	27,648	41.87%	0.70s	1.32s	14.37ms/token	4.36s	477.10s
hit · 08-20 13:46		108	816,288	27,648	44.18%	0.63s	1.32s	14.35ms/token	4.31s	475.05s
c8w4 reuse unique shared 20260820 144339	open comparison									
reuse_blocked_unique_salt · 08-20 14:49		108	816,288	27,648	0.00%	0.91s	1.72s	14.41ms/token	4.58s	507.81s
c8w4 capacity 20260820 151924	open comparison									
block 784 · auto · prefix on · GPU 0.90 · 08-20 15:22		108	816,288	27,648	49.52%	0.63s	1.24s	14.37ms/token	4.30s	472.20s
A_c2_w1 · 08-11 11:19										
fp8	open comparison									
block 1568 · fp8 · prefix on · GPU 0.90 · 08-11 15:22		14	122,460	448	28.17%	1.07s	1.23s	14.61ms/token	1.48s	19.07s
fp16	open comparison									
block 784 · auto · prefix on · GPU 0.90 · 08-11 16:37		14	122,460	448	30.73%	1.03s	1.19s	14.65ms/token	1.49s	18.60s
no cache	open comparison									

arm	comparison	calls	total input	total output	realized reuse	median TTFT	p95 TTFT	median TPOT	median request latency	wall
block 784 · auto · prefix off · GPU 0.90 · 08-12 12:03		14	122,460	448	n/a	1.08s	1.64s	15.99ms/token	1.57s	21.06s
A_c2_w1 · 08-09 21:10										
fp8	open comparison									
block 1568 · fp8 · prefix on · GPU 0.90 · 08-11 16:11		11	96,386	352	19.52%	1.06s	1.24s	14.62ms/token	1.52s	15.33s
fp16	open comparison									
block 784 · auto · prefix on · GPU 0.90 · 08-11 16:30		11	96,386	352	21.15%	1.12s	1.29s	14.60ms/token	1.56s	16.21s
no cache	open comparison									
block 784 · auto · prefix off · GPU 0.90 · 08-12 12:02		11	96,386	352	n/a	1.10s	1.72s	16.02ms/token	1.59s	16.97s
A_c2_w1 · 08-09 19:33										
fp8	open comparison									
block 1568 · fp8 · prefix on · GPU 0.90 · 08-11 16:02		17	165,818	544	29.31%	1.11s	1.40s	14.64ms/token	1.56s	23.92s
fp16	open comparison									
block 784 · auto · prefix on · GPU 0.90 · 08-11 16:34		17	165,818	544	31.68%	1.04s	1.35s	14.64ms/token	1.50s	23.39s
no cache	open comparison									
block 784 · auto · prefix off · GPU 0.90 · 08-12 12:00		17	165,818	544	n/a	1.15s	1.67s	16.01ms/token	1.61s	27.28s
A_c2_w1 · 08-05 15:21										

arm	comparison	calls	total input	total output	realized reuse	median TTFT	p95 TTFT	median TPOT	median request latency	wall
fp16	open comparison									
block 784 · auto · prefix on · GPU 0.90 · 08-11 16:39		12	74,798	384	29.35%	0.69s	1.09s	14.64ms/token	1.14s	14.03s
no cache	open comparison									
block 784 · auto · prefix off · GPU 0.90 · 08-12 11:59		12	74,798	384	n/a	0.74s	1.50s	16.25ms/token	1.24s	17.49s

Summary — tokens

task	reports	vendor	model	calls	total input	median input	total output	cache queried	cache-hit tokens	realized reuse	logical reuse	reuse gap
GenoMAS												
A_c8_w4 · 08-18 15:34	IO · KV	vLLM	Qwen3.6-27B	108	816,288	6,357	319,699	896,418?	281,440	34.48%	52.72%	18.24%
A_c8_w4 · 08-18 14:49	IO · KV	vLLM	Qwen3.6-27B	111	884,081	7,422	211,948	912,832?	349,488	39.53%	51.43%	11.89%
A_c10_w2 · 08-18 13:09	IO · KV	vLLM	Qwen3.6-27B	145	1,194,094	8,593	248,849	n/a	620,272	51.94%	54.92%	2.98%
A_c4_w4 · 08-18 11:35	IO · KV	vLLM	Qwen3.6-27B	36	241,746	6,538	97,849	308,469?	103,008	42.61%	47.25%	4.64%
A_c4_w2 · 08-18 11:35	IO · KV	vLLM	Qwen3.6-27B	53	470,976	9,055	99,201	568,509?	202,736	43.05%	46.69%	3.64%
A_c4_w1 · 08-18 11:35	IO · KV	vLLM	Qwen3.6-27B	59	516,550	9,346	112,804	516,550	192,384	37.24%	41.20%	3.96%
A_c2_w1 · 08-13 09:12	IO · KV	vLLM	Qwen3.6-27B	14	124,099	9,777	28,116	n/a	unmeasured	unmeasured	32.68%	unmeasured
A_c2_w1 · 08-11 11:19	IO · KV	vLLM	Qwen3.6-27B	14	122,460	9,604	28,630	122,460	37,632	30.73%	33.14%	2.41%

task	reports	vendor	model	calls	total input	median input	total output	cache queried	cache-hit tokens	realized reuse	logical reuse	reuse gap
A_c2_w1 · 08-09 21:10	IO · KV	vLLM	Qwen3.6-27B	11	96,386	9,930	28,697	96,386	20,384	21.15%†	23.21%	2.06%†
A_c2_w1 · 08-09 19:33	IO · KV	vLLM	Qwen3.6-27B	17	165,818	9,977	29,720	165,818	71,968	43.40%†	33.57%	-9.83%†
A_c2_w1 · 08-05 15:21	IO · KV	vLLM	Qwen3.6-27B	12	74,798	5,406	29,438	n/a	21,952	29.35%	32.49%	3.14%
A_c4_w2 · 07-29 15:42	IO · KV	FreeInference	qwen3.6-35b	79	837,436	8,709	58,226	n/a	409,600	48.91%	58.48%	9.57%
A_c2_w2 · 07-29 15:34	IO · KV	FreeInference	qwen3.6-35b	36	400,582	11,700	23,287	n/a	204,672	51.09%	59.17%	8.07%
A_c4_w2_nocache · 07-29 13:42	IO · KV	FreeInference	qwen3.6-35b	125	1,708,308	13,308	94,530	n/a	unmeasured	n/a	60.05%	60.05%
A_c2_w2_nocache_openai · 07-29 13:42	IO · KV	FreeInference	qwen3.6-35b	46	764,015	17,299	38,682	n/a	unmeasured	n/a	62.48%	62.48%
A_c2_w2_nocache · 07-29 13:42	IO · KV	OpenAI	gpt-4o-mini-2024-07-18	27	325,139	13,207	5,301	n/a	unmeasured	n/a	66.35%	66.35%
SciLink												
eels_plasmons_basic · 08-12 14:14	IO · KV	vLLM	Qwen3.6-27B	31	224,773	4,418	184,938	152,729?	64,448	42.20%†?	54.18%	11.98%†
SRAgent												
no runs yet												
ChemGraph												
no runs yet												
Montage												
no runs yet												
1000Genome												
no runs yet												
Pi												

task	reports	vendor	model	calls	total input	median input	total output	cache queried	cache-hit tokens	realized reuse	logical reuse	reuse gap
no runs yet												

† cell-level cache-hit token result from the engine's counters. `?` means the counter interval does not match this cell's input, so shared or overlapping server traffic prevents attribution.

`cache queried` is the number of prompt tokens submitted to vLLM's prefix-cache lookup during the measurement window. It is the denominator of the engine's prefix-cache hit rate, not the number of tokens stored in cache.

KV-cache capacity used by these experiments

experiment	vendor · model	capacity	GPU blocks	block size	KV dtype	prefix caching	GPU memory utilization
GenoMAS_3tasks_20260818_113547 / A_c4_w1	vLLM · Qwen3.6-27B	1,325,785 tokens	1,760	784	auto	on	0.9
GenoMAS_3tasks_20260818_113547 / A_c4_w2	vLLM · Qwen3.6-27B	1,325,785 tokens	1,760	784	auto	on	0.9
GenoMAS_3tasks_20260818_113547 / A_c4_w4	vLLM · Qwen3.6-27B	1,325,785 tokens	1,760	784	auto	on	0.9
GenoMAS_A_c10_w2_20260818_130924 / A_c10_w2	vLLM · Qwen3.6-27B	1,325,785 tokens	1,760	784	auto	on	0.9
GenoMAS_A_c2_w1_20260809_211004 / A_c2_w1	vLLM · Qwen3.6-27B	1,325,785 tokens	1,760	784	auto	on	0.9
GenoMAS_A_c2_w1_20260811_111925 / A_c2_w1	vLLM · Qwen3.6-27B	1,325,785 tokens	1,760	784	auto	on	0.9
GenoMAS_A_c2_w1_20260813_091237 / A_c2_w1	vLLM · Qwen3.6-27B	1,349,045 tokens	1,760	784	auto	off	0.9
GenoMAS_A_c8_w4_20260818_144905 / A_c8_w4	vLLM · Qwen3.6-27B	331,446 tokens	440	784	auto	on	0.5
GenoMAS_A_c8_w4_20260818_153414 / A_c8_w4	vLLM · Qwen3.6-27B	207,154 tokens	275	784	auto	on	0.45
SciLink_eels_plasmons_basic_20260812_141445 / eels_plasmons_basic	vLLM · Qwen3.6-27B	1,325,785 tokens	1,760	784	auto	on	0.9

Summary — time

task	reports	vendor	model	calls	total inference	median inference	wall	median TTFT (client)	median TTFT (server)	median queue (server)	median TPOT (client)	median ITL (server)
GenoMAS												
A_c8_w4 · 08-18 15:34	IO · KV	vLLM	Qwen3.6-27B	108	86m15.10s	21.41s	46m15.52s	1.08s	n/a	n/a	15.80ms/token	n/a
A_c8_w4 · 08-18 14:49	IO · KV	vLLM	Qwen3.6-27B	111	59m03.32s	21.73s	17m40.50s	1.14s	n/a	n/a	15.90ms/token	n/a
A_c10_w2 · 08-18 13:09	IO · KV	vLLM	Qwen3.6-27B	145	66m41.32s	21.19s	36m07.55s	0.96s	n/a	n/a	15.10ms/token	n/a
A_c4_w4 · 08-18 11:35	IO · KV	vLLM	Qwen3.6-27B	36	26m12.93s	27.22s	7m44.83s	1.07s	n/a	n/a	15.50ms/token	n/a
A_c4_w2 · 08-18 11:35	IO · KV	vLLM	Qwen3.6-27B	53	27m50.45s	19.88s	14m14.29s	1.09s	n/a	n/a	15.20ms/token	n/a
A_c4_w1 · 08-18 11:35	IO · KV	vLLM	Qwen3.6-27B	59	28m28.11s	19.29s	29m03.15s	1.19s	n/a	n/a	14.30ms/token	n/a
A_c2_w1 · 08-13 09:12	IO · KV	vLLM	Qwen3.6-27B	14	7m08.27s	19.28s	7m13.58s	1.51s	n/a	n/a	14.30ms/token	n/a
A_c2_w1 · 08-11 11:19	IO · KV	vLLM	Qwen3.6-27B	14	7m13.15s	18.07s	7m18.29s	1.40s	n/a	n/a	14.30ms/token	n/a
A_c2_w1 · 08-09 21:10	IO · KV	vLLM	Qwen3.6-27B	11	7m03.88s	18.45s	7m09.03s	1.19s	n/a	n/a	14.30ms/token	n/a
A_c2_w1 · 08-09 19:33	IO · KV	vLLM	Qwen3.6-27B	17	7m20.44s	17.39s	7m30.12s	1.02s	n/a	n/a	14.30ms/token	n/a
A_c2_w1 · 08-05 15:21	IO · KV	vLLM	Qwen3.6-27B	12	7m11.69s	21.54s	7m13.95s	0.73s	n/a	n/a	14.30ms/token	n/a
A_c4_w2 · 07-29 15:42	IO · KV	FreeInference	qwen3.6-35b	79	4m59.84s	1.32s	14m06.72s	0.38s	n/a	n/a	4.30ms/token	n/a
A_c2_w2 · 07-29 15:34	IO · KV	FreeInference	qwen3.6-35b	36	1m53.24s	1.50s	4m34.55s	0.38s	n/a	n/a	4.10ms/token	n/a
A_c4_w2_nocache · 07-29 13:42	IO · KV	FreeInference	qwen3.6-35b	125	9m22.54s	1.76s	8m41.42s	0.71s	n/a	n/a	4.60ms/token	n/a

task	reports	vendor	model	calls	total inference	median inference	wall	median TTFT (client)	median TTFT (server)	median queue (server)	median TPOT (client)	median ITL (server)
A_c2_w2_nocache_openai · 07-29 13:42	IO · KV	FreeInference	qwen3.6-35b	46	4m28.38s	3.56s	7m14.15s	0.91s	n/a	n/a	5.30ms/token	n/a
A_c2_w2_nocache · 07-29 13:42	IO · KV	OpenAI	gpt-4o-mini-2024-07-18	27	3m54.11s	5.21s	4m21.17s	1.66s	n/a	n/a	29.10ms/token	n/a
SciLink												
eels_plasmons_basic · 08-12 14:14	IO · KV	vLLM	Qwen3.6-27B	31	46m38.45s	56.03s	31m58.67s	1.05s	n/a	n/a	14.40ms/token	n/a
SRAgent												
no runs yet												
ChemGraph												
no runs yet												
Montage												
no runs yet												
1000Genome												
no runs yet												
Pi												
no runs yet												